Disaster Response Coordination Hub — Final SRS (Revised)

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IT314 Software Engineering · AI-Assisted Incident Triage, Dispatch and Inter-Agency Coordination. Revised to a fixed scope: **three roles** (Normal User, Disaster Management, Application Management) and **three scenarios** (Cyclone, Industrial Fire, Urban Flooding). Agencies are an external integration with no login. Web platform only.

Core baseline: 42 FRs, 42 NFRs, 25 error cases, 10 execution specifications, 11 scenario examples, 12 workflow steps. Items marked *(Optional)* are extensions and are excluded from the core counts.

1. Project Overview

The Disaster Response Coordination Hub is a role-based **web platform** that connects citizens, authorised disaster-management personnel and application administrators. It captures incident information, supports human-led verification and AI-assisted triage, coordinates resources, manages dispatch, publishes verified updates, and tracks monetary and physical aid.

The system supports **only Cyclone, Industrial Fire and Urban Flooding**. The scenario is a fixed choice, offered as a closed list on the report form and validated again on the server; any other value is rejected.

The system has **exactly three roles**: Normal User, Disaster Management and Application Management. Publisher/Approver, moderator, closer and supervisor are *permissions inside Disaster Management*, not separate roles. External agencies are an integration, not a login type: Disaster Management staff record an agency's acknowledgement, or a trusted integration reports it.

AI is advisory. It may suggest classification, urgency factors and duplicate candidates, but it never replaces an authorised human decision.

Monetary donations and physical-aid contributions are independent entry points. A user may contribute without filing an incident. Payment, dispatch, notification and delivery outcomes are never presented as successful without confirmation.

Supported scenario focus

Scenario	Operational focus	Typical captured information
Cyclone	Warnings, affected areas, shelter and support needs	Location, severity, affected people, evidence, assistance need
Industrial Fire	Fire/smoke verification and emergency coordination	Site, smoke/fire indicators, injuries, access constraints, evidence
Urban Flooding	Stranded persons, water accumulation, access and rescue coordination	Location, water level, people affected, road/access status, evidence

2. Scope, Assumptions and Stakeholders

In scope: citizen reporting and contributions; operational verification, evidence moderation, AI-assisted triage, prioritisation, mapping, resource coordination, dispatch and inter-agency messaging; incident closure; safety-content and campaign management; verified public updates; abuse and rate-limit controls; administration, security, monitoring, audit, backup/recovery and payment/notification integration.

Out of scope (recommended exclusions): additional roles beyond the three; other disaster types; unauthenticated (anonymous) reporting; agency or partner login accounts; autonomous dispatch; unverified claims presented as official facts; uncontrolled payment handling; and AI-only operational decisions.

Stakeholders

Stakeholder	Responsibilities	Authority boundary
Normal User	Registration/login, profile, incident reporting, evidence, tracking/history, responding to information requests, withdrawing or updating own reports, alerts, safety information, monetary donations and physical aid.	Own permitted records and contributions; no verification, dispatch or admin authority.

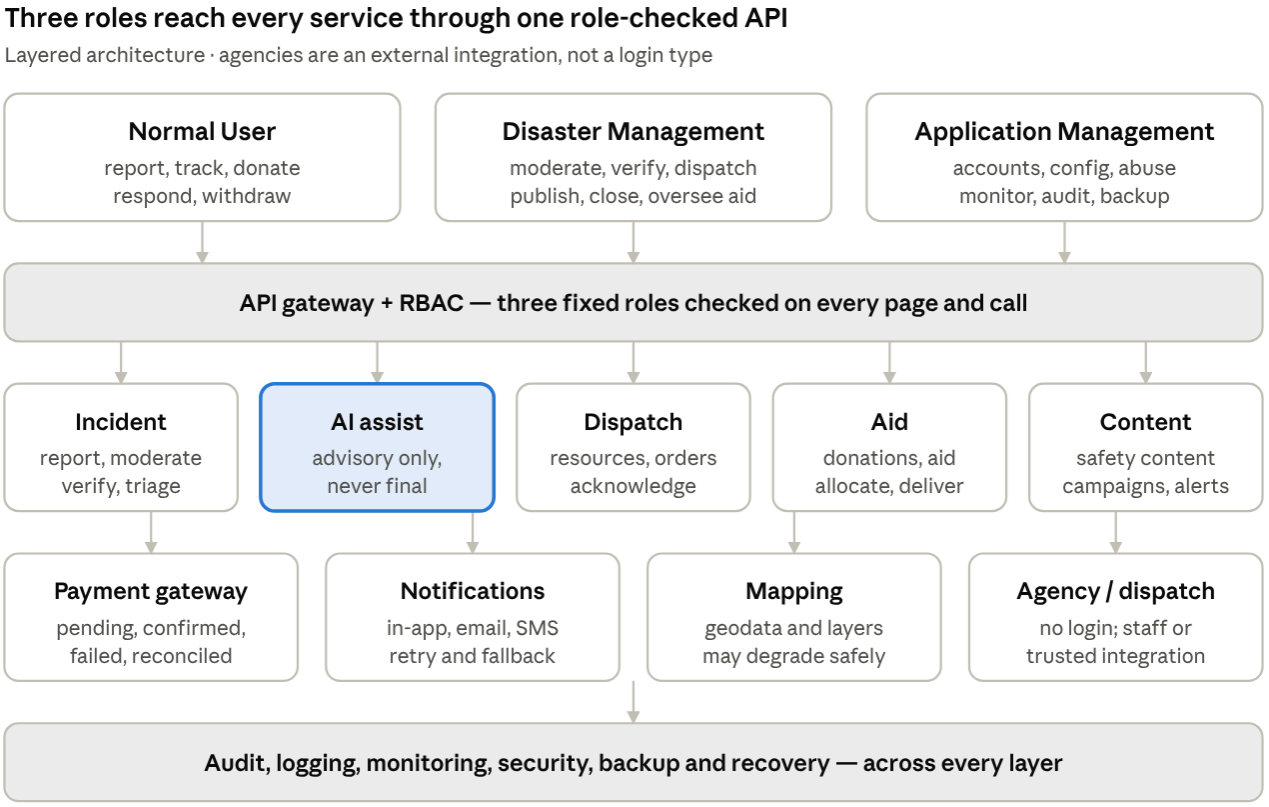
Stakeholder	Responsibilities	Authority boundary
Disaster Management	Staff login, evidence moderation, verification, duplicate review, AI-assisted classification/triage, priority, mapping, resources, dispatch, agency messaging, incident closure, safety content, campaigns, verified updates, donation/aid oversight, allocation and analytics.	Authorised operational decisions; human accountability for critical actions.
Application Management	Admin login, accounts/roles, configuration, abuse and rate-limit controls, moderation configuration, monitoring, audit, data, security, backup/recovery, notifications and payment integration monitoring.	Platform administration; changes must remain controlled and auditable.

Note. Publisher/Approver, moderator, closer and supervisor are **permissions inside Disaster Management**, not additional stakeholder roles.

Assumptions

- Users access the platform through supported desktop and mobile browsers.
- GPS may be unavailable, so manual location entry is supported.
- External payment, map, notification and agency/dispatch services may fail.
- Evidence may require security scanning and human moderation before it is shown.
- Resource and dispatch confirmation comes from authorised operational actors or trusted integrations.
- **Agencies are external parties.** They hold no accounts. Disaster Management staff record an agency's acknowledgement or rejection, or a trusted integration reports it.
- Spam, abusive and bulk-automated submissions are expected and must be controlled.
- Any measurable performance or availability target is a **proposed target until approved by [named approver]**.
- *(Optional)* Intermittent connectivity handling is limited to saving an unsent report draft in the browser.
- *(Optional)* Multilingual content is not part of this baseline and would require a separate change.

3. System Architecture and Response Principles



layered architecture · 3 actors, 5 services, 4 external integrations

Every request passes the same API and RBAC layer, so a page or call outside a role's permissions is denied and logged. AI assistance sits inside the service layer and produces recommendations only. Agencies sit in the external row: they receive dispatch messages but hold no account, and their responses re-enter the system through staff entry or a trusted integration.

Core principles

Principle	Required behaviour
Human oversight	AI recommendations are labelled advisory; authorised staff accept, modify or reject them.
State transparency	Explicit states are used throughout: Submitted, Pending Moderation, Approved, Quarantined, Needs Information, Verified, Rejected, Withdrawn, Out of Scope, Pending, Confirmed, Failed, Allocated, Requested, Acknowledged, Dispatched, Escalated, Delivered and Closed.

Principle	Required behaviour
No false success	Payment, dispatch, notification and aid delivery require the relevant confirmation event.
Auditability	Critical actions record actor, timestamp, affected record, action and result; important decisions include rationale.
Least privilege	RBAC restricts data and actions by role; permissions inside Disaster Management are granted separately.
Fixed scope	Three roles and three scenarios. Neither list can be extended through configuration; both need a formal change.
Abuse resistance	Public endpoints are rate-limited and abuse-controlled; suspected abuse is contained, not silently dropped.
Content safety	User text and uploaded files are never rendered unsafely, and evidence is moderated before it is shown publicly.
External by integration	Agencies, payment, mapping and notification providers are integrations. None of them is a login type.

Response loop

Capture → Validate → Store and acknowledge → Moderate → Review → AI assist → Human decision → Act → Update and monitor → Close.

(Optional) A dedicated state-machine section (Section 16) may be added later for incident, dispatch, donation and aid states; it is not part of this baseline.

4. Functional Requirements – Normal User

Normal users access reporting, information, tracking and contribution workflows. Monetary donations and physical-aid contributions are mandatory features and may be used without reporting an incident. **12 core requirements; 15 with the optional items.**

NU-FR-01 / Registration and Login

Field	Specification
Description	Create an account and authenticate securely. Public sign-up creates **Normal Users only**; staff and admin accounts are created only by Application Management.
Input	Credentials, profile, consent
Internal processing	Validate; protect credentials; create account/session; assign the Normal User role; enforce RBAC; apply rate limits (AM-FR-11).
Expected output	Account and session
Application reaction	Show the result; dashboard only after authentication.
Success	Valid account and session
Failure	Duplicate account; invalid credentials; identity-service failure; rate limit reached

NU-FR-02 / Profile Management

Field	Specification
Description	View and update permitted profile data.
Input	Profile changes
Internal processing	Authenticate; validate; persist; audit where required. A user cannot change their own role.
Expected output	Updated profile
Application reaction	Show saved values; reject invalid fields.
Success	Change committed
Failure	Invalid data; stale session; database failure; attempted self-escalation

NU-FR-03 / Incident Reporting

Field	Specification
Description	Report an incident. The scenario is a **fixed choice of the three supported types** (Cyclone, Industrial Fire, Urban Flooding).
Input	Scenario, description, urgency indicators, location, people affected
Internal processing	Validate; **the server rejects any scenario value outside the three**; generate ID; timestamp; set Submitted; queue moderation and review; evaluate duplicate candidates.
Expected output	Incident record with a unique ID
Application reaction	Show the ID and Submitted status; never claim verification or dispatch.
Success	Stored with acknowledgement
Failure	Missing data; unsupported scenario (EH-25); duplicate suspicion; server failure

NU-FR-04 / Location and Details

Field	Specification
Description	Capture a GPS/map point or a manual location and the relevant details.
Input	Coordinates, address, details
Internal processing	Validate and normalise; store source and accuracy.
Expected output	Location-linked details
Application reaction	Show a map or address preview; allow correction.
Success	Correct location attached
Failure	GPS unavailable; invalid or ambiguous location; map service outage

NU-FR-05 / Evidence Upload

Field	Specification
Description	Attach permitted evidence to a report.
Input	Files and incident ID
Internal processing	Validate type, size and safety; scan; store; link; mark Pending Moderation where configured (DM-FR-18).
Expected output	Evidence reference and moderation state
Application reaction	Show per-file state; reject unsafe files; never render unchecked content.
Success	Stored and linked
Failure	Unsupported, corrupt or malicious file; upload timeout

NU-FR-06 / Report Tracking, History and Receipts

Field	Specification
Description	View own authorised reports, timelines, contribution history and confirmed donation receipts.
Input	User session and filters
Internal processing	Query permitted records only; build timeline; retrieve receipts for confirmed transactions.
Expected output	History, status and receipts
Application reaction	Show the current state, verified updates and closure reason where applicable.
Success	Accurate authorised history
Failure	Access denied; missing record; database failure

NU-FR-07 / Alerts and Notifications

Field	Specification
Description	Receive relevant official alerts and transaction or report updates.

Field	Specification
Input	Event and user preferences, including opt-out where offered
Internal processing	Check eligibility; queue and send; record the provider result; retry per policy.
Expected output	Queued / sent / delivered / failed state
Application reaction	Show in-app status, which is the source of truth; retry transient failures.
Success	Provider accepts, or an in-app status exists
Failure	Provider outage; invalid contact; rate limit

NU-FR-08 / Safety Information

Field	Specification
Description	Present approved safety guidance for the three supported scenarios.
Input	Scenario, location, context
Internal processing	Retrieve approved, versioned content published under DM-FR-17.
Expected output	Safety guidance
Application reaction	Show official status, source and last update time.
Success	Approved content visible
Failure	Content service outage; stale or unapproved content withheld

NU-FR-09 / Monetary Donations

Field	Specification
Description	Donate money securely, independently of incident reporting.
Input	Amount, purpose or campaign, donor information, payment method

Field	Specification
Internal processing	Create Pending; secure gateway handoff; validate the signed callback; reconcile; issue a receipt only after confirmation; idempotency prevents duplicates.
Expected output	Pending / Confirmed / Failed, plus a receipt if confirmed
Application reaction	Show the exact payment state; never success without confirmation.
Success	Gateway-confirmed transaction
Failure	Decline; gateway outage; callback delay; duplicate risk

NU-FR-10 / Physical Aid Contribution

Field	Specification
Description	Register physical aid independently of incident reporting.
Input	Aid type, quantity, condition, location, contact
Internal processing	Registered → verification → allocation → delivery tracking.
Expected output	Contribution record and status
Application reaction	Show each stage separately; no delivery success without confirmation.
Success	Verified, allocated and delivery confirmed
Failure	Invalid or unavailable aid; no matching need; delivery failure

NU-FR-11 / Respond to Information Requests

Field	Specification
Description	Answer a staff request for more information on an own report that is in the Needs Information state.

Field	Specification
Input	Incident ID, additional description, evidence or corrected location
Internal processing	Verify ownership; append the response with actor and timestamp; return the incident to the review queue.
Expected output	Updated incident with a response entry
Application reaction	Show the request, the response and the resulting state change.
Success	Response recorded and the incident re-queued for review
Failure	Not the owner; incident already closed; response window expired

NU-FR-12 / Withdraw or Update Own Report

Field	Specification
Description	Correct or withdraw an own report before an operational action depends on it.
Input	Incident ID, correction or withdrawal reason
Internal processing	Verify ownership and current state; block withdrawal once dispatch has started; preserve the original values in history.
Expected output	Updated or Withdrawn incident
Application reaction	Show the new state and that staff have been informed; nothing is deleted.
Success	Change committed with an auditable history entry
Failure	Not the owner; state no longer permits change; concurrent staff update

NU-FR-13 / Browser Draft and Safe Retry *(Optional)*

Field	Specification
Description	Save an unsent report draft in the browser and retry submission safely.
Input	Partially completed report
Internal processing	Store the draft locally; on retry use an idempotency key so no duplicate incident is created.
Expected output	Draft marked "Not yet sent"
Application reaction	Make clear that a draft is not a submitted report.
Success	Draft restored, or submitted exactly once
Failure	Local storage unavailable; draft expired; duplicate submission blocked

NU-FR-14 / Data Export and Deletion Request *(Optional)*

Field	Specification
Description	Request an export or deletion of own personal data.
Input	Authenticated request and request type
Internal processing	Verify identity; queue for oversight under AM-FR-13; retain records required for audit or finance.
Expected output	Request state and outcome
Application reaction	Show what can and cannot be removed, and why.
Success	Request completed and recorded
Failure	Identity not verified; retention obligation prevents deletion; export generation failure

NU-FR-15 / Flag Published or Shared Content *(Optional)*

Field	Specification
Description	Report content that appears unsafe, abusive or incorrect.
Input	Content reference and reason

Field	Specification
Internal processing	Rate-limit the flag; queue it for moderation under DM-FR-18 and AM-FR-12.
Expected output	Flag record
Application reaction	Confirm receipt without revealing moderation outcomes.
Success	Flag queued for review
Failure	Duplicate or abusive flagging; content already removed

5. Functional Requirements – Disaster Management

Authorised operational staff control moderation, verification, triage, prioritisation, resource coordination, dispatch, agency messaging, public content, closure and aid oversight. AI remains advisory. Publisher/Approver, moderator, closer and supervisor are permissions held inside this single role. **18 core requirements; 20 with the optional items.**

DM-FR-01 / Staff Login

Field	Specification
Description	Authenticate authorised staff.
Input	Credentials and MFA
Internal processing	Authenticate; apply RBAC and the staff member's permissions; create a secure session.
Expected output	Operational session
Application reaction	Load the dashboard only after authentication.
Success	Authorised session with the correct permission set
Failure	Invalid or expired account; MFA failure; suspended account

DM-FR-02 / Incident Dashboard

Field	Specification
Description	Consolidated incidents, priorities, maps, moderation queue and pending actions.
Input	Filters
Internal processing	Query and aggregate authorised data; refresh; surface overdue dispatches.
Expected output	Dashboard
Application reaction	Show status and last-updated time.
Success	Accurate current data
Failure	Stale data; query or integration failure

DM-FR-03 / Review and Verification

Field	Specification
Description	Review evidence and set the authorised verification state.
Input	Incident, evidence, decision, rationale
Internal processing	Inspect; save decision, actor and time; publish only an approved status.
Expected output	Verification state
Application reaction	Show the decision; notify the reporter where appropriate.
Success	Auditable decision saved
Failure	Insufficient evidence; concurrent update; unauthorised action

DM-FR-04 / Duplicate Detection

Field	Specification
Description	Surface likely duplicate reports without deleting them automatically.
Input	Text, location, time, evidence

Field	Specification
Internal processing	Rules and AI similarity; present candidates; preserve originals.
Expected output	Duplicate candidates
Application reaction	Show matches and rationale; a human decides.
Success	Relationship recorded
Failure	False match; uncertain similarity; service outage

DM-FR-05 / AI Classification and Triage

Field	Specification
Description	Suggest scenario, urgency factors and triage evidence.
Input	Report, evidence, location, context
Internal processing	AI inference with confidence and uncertainty; guardrails; store the recommendation separately from the human decision. **A suggestion outside the three supported scenarios is flagged Out of Scope and staff triage it manually.**
Expected output	Advisory recommendation
Application reaction	Label AI output clearly; the human accepts, modifies or rejects it.
Success	Human final decision recorded
Failure	Low confidence; missing data; model outage or unsafe output

DM-FR-06 / Prioritisation

Field	Specification
Description	Assign operational priority using facts and approved guidance.
Input	Incident, indicators, AI advice
Internal processing	Rules plus human judgement; save the rationale.

Field	Specification
Expected output	Priority
Application reaction	Update the queue, map and resource planning.
Success	Authorised priority saved
Failure	Insufficient or conflicting information

DM-FR-07 / Mapping

Field	Specification
Description	Map incidents, resources, routes and relevant areas.
Input	Geodata and layers
Internal processing	Validate coordinates; render map layers.
Expected output	Operational map
Application reaction	Show pins, status and data freshness.
Success	Map reflects authoritative data
Failure	Map outage; inaccurate location

DM-FR-08 / Resource Coordination

Field	Specification
Description	Track and assign teams, vehicles, supplies and support resources.
Input	Resource data and assignment
Internal processing	Check availability; reserve and assign; prevent conflicts.
Expected output	Resource assignment
Application reaction	Show availability and conflicts.
Success	Assignment committed
Failure	Unavailable, conflicting or stale resource

DM-FR-09 / Dispatch

Field	Specification
Description	Create and track an authorised emergency dispatch.
Input	Verified and prioritised incident plus a resource
Internal processing	Check prerequisites; create the order; transmit; await acknowledgement; start the configured timeout and escalation.
Expected output	Dispatch state
Application reaction	Show Requested / Acknowledged / Dispatched / Rejected / Escalated; never a false success.
Success	Receiving confirmation recorded
Failure	No resource; rejection; transmission outage; cancellation; timeout

DM-FR-10 / Inter-Agency Communication

Field	Specification
Description	Share authorised coordination messages with **external agency contacts**. Agencies hold no accounts.
Input	Message, recipients, context
Internal processing	Check authority; route through the messaging or dispatch integration; log delivery. **Staff record the agency's acknowledgement or rejection, or a trusted integration reports it**, with the timestamp and the recorder stored.
Expected output	Message and acknowledgement state
Application reaction	Show sent / acknowledged / rejected / failed, and who recorded it.
Success	Message accepted, and any acknowledgement attributed to its recorder
Failure	Unauthorised recipient; provider failure; no acknowledgement received

DM-FR-11 / Verified Public Updates

Field	Specification
Description	Publish authoritative public updates. Requires the **Publisher/Approver permission** within Disaster Management.
Input	Approved text, status, area
Internal processing	Validate the publishing permission; version; publish; audit.
Expected output	Verified update
Application reaction	Timestamped update with an optional notification.
Success	Visible through an approved channel
Failure	Missing publishing permission; invalid content; notification failure

DM-FR-12 / Donation and Aid Oversight

Field	Specification
Description	Monitor confirmed money and verified aid.
Input	Contribution, campaign and need data
Internal processing	Aggregate confirmed and verified states; maintain traceability.
Expected output	Oversight view
Application reaction	Distinguish pending, confirmed and verified clearly.
Success	Reconciled view
Failure	Gateway delay; inconsistency

DM-FR-13 / Aid Allocation

Field	Specification
Description	Allocate verified aid to an approved need and track delivery. Needs come from DM-FR-19 where adopted, otherwise they are entered directly by staff here.
Input	Verified aid and a need
Internal processing	Check compatibility and quantity; reserve; create a delivery task.
Expected output	Allocation and delivery state
Application reaction	Show Allocated / In Transit / Delivered / Failed.
Success	Delivery confirmed
Failure	No match; insufficient quantity; delivery failure

DM-FR-14 / Analytics

Field	Specification
Description	Provide operational and contribution analytics.
Input	Filters: time, scenario, region
Internal processing	Aggregate authorised data; distinguish unknown and pending; count Out of Scope records separately.
Expected output	Charts and tables
Application reaction	Show definitions and data freshness.
Success	Consistent analytics
Failure	Incomplete data; query failure

DM-FR-15 / Incident Closure

Field	Specification
Description	Close an incident with a recorded reason. Requires the closure permission.
Input	Incident, closure reason, optional public summary

Field	Specification
Internal processing	Check that no dispatch or allocation is still open, or require an explicit override with rationale; set Closed; freeze further edits; keep the record readable.
Expected output	Closed incident with reason
Application reaction	Show the closure reason and time to staff and to the reporter.
Success	Closed with an auditable reason
Failure	Open dependent actions; missing permission; missing reason

DM-FR-16 / Campaign Management

Field	Specification
Description	Create and manage approved donation purposes and campaigns.
Input	Campaign name, purpose, scenario, period, target
Internal processing	Validate; **the campaign scenario field accepts only the three supported scenarios**; version; activate or close; audit.
Expected output	Active or closed campaign
Application reaction	Show only active, approved campaigns as donation purposes.
Success	Campaign active and selectable
Failure	Invalid scenario; duplicate campaign; closed campaign still referenced

DM-FR-17 / Safety Content Authoring and Approval

Field	Specification
Description	Author, version and approve safety guidance, **versioned per scenario**.
Input	Draft content, scenario, source

Field	Specification
Internal processing	Draft → review → approval by a holder of the Publisher/Approver permission → publish; keep every version.
Expected output	Approved, versioned content
Application reaction	Only approved versions are shown under NU-FR-08.
Success	Approved version published and attributable
Failure	Unapproved draft; missing permission; superseded version still served

DM-FR-18 / Evidence and Content Moderation

Field	Specification
Description	Review uploaded evidence and user text before it is shown beyond the reviewing staff.
Input	Evidence item, flag, moderation decision
Internal processing	Apply the moderation policy configured in AM-FR-12; set Approved, Quarantined or Removed; record actor, time and reason.
Expected output	Moderation state
Application reaction	Quarantined content is never rendered publicly; staff see a placeholder with the reason.
Success	Decision recorded and enforced
Failure	Backlog beyond the configured threshold; unsafe content detected; unauthorised moderation attempt

DM-FR-19 / Needs Management *(Optional)*

Field	Specification
Description	Record, prioritise and close operational needs that aid can be allocated against.

Field	Specification
Input	Need type, quantity, location, priority, linked incident
Internal processing	Validate; deduplicate; track fulfilled quantity against the requirement.
Expected output	Need record and fulfilment state
Application reaction	Show open, partially met and met needs.
Success	Need recorded and traceable to allocations
Failure	Duplicate need; over-allocation; stale need

DM-FR-20 / Agency Directory *(Optional)*

Field	Specification
Description	Maintain a directory of **external agency contacts** and assign them to incidents. **No accounts are created and no login is granted.**
Input	Agency name, contact channel, coverage area, assignment
Internal processing	Validate the contact; link to incidents; use the directory as the recipient list for DM-FR-10.
Expected output	Agency contact records and assignments
Application reaction	Show which contacts are assigned to an incident and through which channel.
Success	Contact assigned and usable as a dispatch recipient
Failure	Invalid contact; unreachable channel; duplicate entry

6. Functional Requirements – Application Management

Administrators maintain platform accounts, configuration, abuse controls, moderation policy, observability, audit, security, recovery and external integrations. There is **no agency account management**, because agencies are an external integration. **12 core requirements; 14 with the optional items.**

AM-FR-01 / Administrator Login

Field	Specification
Description	Authenticate administrators.
Input	Admin credentials and MFA
Internal processing	Authenticate; apply RBAC; create a secure session.
Expected output	Admin session
Application reaction	Load the console only after authentication.
Success	Authorised session
Failure	Invalid or disabled account; MFA failure

AM-FR-02 / Account and Role Management

Field	Specification
Description	Manage accounts and permissions. **Only three roles exist: Normal User, Disaster Management and Application Management.** Staff and admin accounts are created only here; public sign-up creates Normal Users only. Publisher/Approver, moderator, closer and supervisor are granted as permissions inside Disaster Management.
Input	Account or role request, permission grant, suspension or reactivation
Internal processing	Validate authority; apply RBAC; reject any role outside the three; prevent self-escalation; prevent removal of the last active administrator; audit before and after.
Expected output	Account, role and permission state
Application reaction	Show the effective role, the granted permissions and the audit result.
Success	Committed change
Failure	Insufficient privilege; invalid role; self-escalation attempt; last-admin protection triggered

AM-FR-03 / Configuration Management

Field	Specification
Description	Manage approved scenario, notification and operational settings.
Input	Configuration change
Internal processing	Validate the schema; version; apply; audit before and after. **The scenario setting can only enable or disable the three supported scenarios; it cannot add a fourth.** Also covers the dispatch timeout and escalation values and the duplicate-detection switch.
Expected output	Active configuration version
Application reaction	Show the effective configuration and its history.
Success	Valid configuration active
Failure	Invalid or unsafe change; attempt to add a scenario or role; service failure

AM-FR-04 / System Monitoring

Field	Specification
Description	Monitor services, queues, storage and integrations, including the moderation backlog and the agency/dispatch integration.
Input	Telemetry and events
Internal processing	Collect metrics and logs; evaluate health; raise alerts.
Expected output	Health dashboard
Application reaction	Show the current or last-known state and open incidents.
Success	Accurate telemetry
Failure	Monitoring blind spot or outage

AM-FR-05 / Logging and Audit

Field	Specification
Description	Maintain security and operational audit records, including moderation decisions, closures and **staff-recorded agency acknowledgements**.
Input	Events
Internal processing	Normalise; timestamp; protect integrity; retain per policy.
Expected output	Audit log
Application reaction	Search and filter under access controls.
Success	Retrievable, protected log
Failure	Logging outage; integrity failure

AM-FR-06 / Data Management

Field	Specification
Description	Control validation, correction, lifecycle and retention.
Input	Data-quality or correction task
Internal processing	Validate authority; preserve traceability; apply controlled change.
Expected output	Data state
Application reaction	Show the operation result and affected records.
Success	Controlled change complete
Failure	Integrity conflict; unauthorised bulk change

AM-FR-07 / Security Management

Field	Specification
Description	Manage security controls, secrets and incidents.
Input	Security configuration or event

Field	Specification
Internal processing	Apply policy; protect and rotate secrets; monitor and respond.
Expected output	Security state
Application reaction	Show actionable alerts; audit every change.
Success	Control applied and logged
Failure	Misconfiguration; secret failure

AM-FR-08 / Backup and Recovery

Field	Specification
Description	Create, validate and restore backups.
Input	Backup or restore request
Internal processing	Protected backup; integrity check; safe restore.
Expected output	Backup and recovery state
Application reaction	Show backup age, status and restore verification.
Success	Restored data verified
Failure	Corrupt backup; storage failure; restore conflict

AM-FR-09 / Notification Configuration

Field	Specification
Description	Configure channels, templates, routing and retry, including the in-app fallback channel.
Input	Notification configuration
Internal processing	Validate, version, test and activate.
Expected output	Active notification configuration
Application reaction	Preview and test, with the version shown.
Success	Valid active configuration
Failure	Provider or template incompatibility

AM-FR-10 / Payment Integration Monitoring

Field	Specification
Description	Monitor the gateway, callbacks and reconciliation.
Input	Gateway health and callbacks
Internal processing	Verify callbacks; reconcile; flag timeouts and mismatches.
Expected output	Integration health and exceptions
Application reaction	Show Pending, Confirmed, Failed and mismatches distinctly.
Success	Transactions reconciled or safely held
Failure	Gateway outage; invalid callback; mismatch

AM-FR-11 / Abuse and Rate-Limit Controls

Field	Specification
Description	Protect public endpoints against spam, bulk automation and abusive submissions.
Input	Request patterns, thresholds, block and allow lists
Internal processing	Apply per-account, per-IP and per-endpoint limits to sign-up, login, reporting, uploads, flags and donations; throttle, challenge or block; log every action.
Expected output	Limit decision and abuse record
Application reaction	Explain that a limit was reached without revealing the thresholds; never silently discard a legitimate report.
Success	Abuse contained with the legitimate path preserved
Failure	Limits too strict and blocking real reports; distributed abuse below the threshold; control service outage

AM-FR-12 / Moderation Configuration

Field	Specification
Description	Configure the moderation policy that DM-FR-18 enforces.
Input	Allowed file types, scan settings, auto-quarantine rules, backlog thresholds
Internal processing	Validate; version; apply; audit; alert when the backlog exceeds its threshold.
Expected output	Active moderation policy
Application reaction	Show the effective policy and its version to moderators.
Success	Policy active and enforced
Failure	Unsafe policy; scanner unavailable; backlog threshold breached

AM-FR-13 / Data-Request Oversight *(Optional)*

Field	Specification
Description	Oversee export and deletion requests raised under NU-FR-14.
Input	Request queue and retention rules
Internal processing	Verify identity checks; apply retention and legal holds; execute or refuse with a recorded reason.
Expected output	Request outcome
Application reaction	Show what was exported, removed or retained, and why.
Success	Request resolved and auditable
Failure	Retention conflict; incomplete export; unverified requester

AM-FR-14 / Configuration Approval *(Optional)*

Field	Specification
Description	Require a second approver for critical configuration and role changes.
Input	Proposed change and approver decision
Internal processing	Hold the change pending; require a different administrator to approve; apply only after approval; audit both actors.
Expected output	Approved or rejected change
Application reaction	Show pending changes and who must approve them.
Success	Critical change applied with two recorded approvers
Failure	Only one administrator available; approval timeout; self-approval attempt

7. Detailed Functionality Execution Specifications

Every functionality follows the same safety pattern: preconditions → user steps → input validation → internal processing → persisted state → application response → notifications → confirmed success or controlled exception. **10 core specifications; 11 with the optional item.**

7.1 Incident reporting

Aspect	Execution detail
Purpose	Capture a Cyclone, Industrial Fire or Urban Flooding incident for authorised review.
Stakeholder	Normal User; Disaster Management subsequently moderates, verifies and triages.
Preconditions	Authenticated user; one of the three supported scenarios selected; minimum required details available; the account is within its rate limit (AM-FR-11).

Aspect	Execution detail
User steps	Report Incident → select scenario → enter location and details → add evidence → review → submit.
Input	Scenario, location, description, urgency indicators, people affected, evidence.
Internal processing	Validate; **the scenario must be one of the three, validated on the server** regardless of what the form sent; generate a unique ID; timestamp; store; set Submitted; queue moderation, audit and review events; evaluate duplicate candidates.
Application response	Show the ID and Submitted status, and state explicitly that submission is neither verification nor dispatch.
Expected screen output	Incident details, status timeline, evidence and moderation state, next-step guidance.
Database and status changes	Incident created; status = Submitted; evidence linked and marked Pending Moderation; notification event queued.
Notifications	User acknowledgement and staff queue notification per configuration.
Success	Stored incident with a unique ID and an acknowledgement.
Failure	Highlight invalid fields; isolate upload failures; reject an unsupported scenario (EH-25); preserve a safe retry; never show a false submission success.
Flow	Submit → validate → store as Submitted → moderate evidence → staff review → Verified / Needs Information / Rejected → AI assist → human triage → dispatch if authorised.
Cross-references	NU-FR-03, NU-FR-04, NU-FR-05, DM-FR-18, DM-FR-03, **AM-FR-11**.

7.2 AI-assisted triage

Aspect	Execution detail
Purpose	Surface the likely scenario, urgency indicators, duplicate candidates and relevant evidence to reduce staff workload.
Preconditions	Incident exists; authorised staff; AI service available, or the manual fallback is available.
User steps	Open incident → view or request AI assistance → inspect factors and uncertainty → accept, modify or reject → record the final decision.
Internal processing	AI analyses permitted inputs and returns a recommendation with uncertainty; guardrails check the output; the recommendation is stored separately from the human decision. **A scenario outside the three is flagged Out of Scope and staff decide.**
Application response	The advisory panel clearly labels AI output. Low-confidence and missing-data cases are highlighted, and the manual workflow remains available.
Database and status changes	AI recommendation with model and version metadata; the human classification, priority and verification are stored separately.
Success	An authorised human records the final decision after considering the evidence and the AI assistance.
Failure	AI unavailable or uncertain → manual triage continues; no operational action is blocked solely by an AI outage.
Flow	Incident → AI recommendation → explain factors → human review → accept / modify / reject → authoritative state.
Cross-references	DM-FR-05, DM-FR-04, DM-FR-06, DM-NFR-09, DM-NFR-10.

7.3 Monetary donation

Aspect	Execution detail
Purpose	Accept monetary contributions securely to an approved purpose or campaign, independently of incident reporting.
Preconditions	An approved, active campaign exists (DM-FR-16, **limited to the three scenarios**); the payment integration is available; the user has a valid payment method; the request is within its rate limit (**AM-FR-11**).
User steps	Donate → choose purpose → enter amount → review → secure gateway → authenticate → return to the application.
Internal processing	Create a Pending transaction → secure gateway handoff → validate the signed callback or status → reconcile → Confirmed / Failed / Pending → receipt only after confirmation.
Application response	Pending while processing; Confirmed only after gateway verification; Failed on explicit rejection; Pending on an unknown or timed-out state.
Database and status changes	Transaction state machine plus audit and reconciliation records; idempotency prevents accidental duplicates.
Notifications	Receipt and confirmation after a confirmed payment; failure or pending notice per configuration.
Success	The gateway confirms and the application verifies the transaction reference or callback.
Failure	Decline → Failed; outage or timeout → Pending or temporarily unavailable; no false receipt; a safe retry and reconciliation path exists.
Flow	Donation form → Pending → secure gateway → gateway result → verify signature and reference → Confirmed / Failed / Pending → receipt only if Confirmed.

Aspect	Execution detail
Cross-references	NU-FR-09, DM-FR-16, DM-FR-12, AM-FR-10, **AM-FR-11**.

7.4 Physical aid contribution

Aspect	Execution detail
Purpose	Register physical aid and track verification, allocation and delivery, independently of incident reporting. Needs come from DM-FR-19 where adopted; otherwise **Disaster Management staff enter needs directly under DM-FR-13**.
Preconditions	Supported aid type; the contributor can state quantity, condition and location; the request is within its rate limit (**AM-FR-11**).
User steps	Contribute Aid → select type → enter quantity, condition, availability and location → submit → await verification → follow allocation and delivery.
Internal processing	Registered / Pending Verification → staff verify → Verified or Rejected → Allocate → delivery task → In Transit → Delivered or Failed.
Application response	Each stage is shown separately; no delivery success before an authorised confirmation.
Database and status changes	Contribution, verification, allocation and delivery events persist with an audit history.
Notifications	Verification result, allocation instructions, pickup, transit and delivery updates per configuration.
Success	Aid verified, allocated to a suitable need, and delivery confirmed by **Disaster Management staff or a trusted integration**.
Failure	Invalid or unavailable aid, or no matching need → an appropriate non-success state with a recovery option.
Flow	Register → Pending Verification → Verified → Allocated → In Transit → Delivered, or a recorded non-success state.

Aspect	Execution detail
Cross-references	NU-FR-10, DM-FR-13, DM-FR-12, DM-FR-19 *(optional)*.

7.5 Emergency dispatch

Aspect	Execution detail
Purpose	Move from a verified and prioritised incident to controlled emergency resource deployment, **only for incidents of the three supported scenarios**.
Preconditions	Incident sufficiently verified; priority assigned; resource available; the staff member holds dispatch authority; the **receiving team or agency contact is listed in DM-FR-10** (or in the DM-FR-20 directory, if adopted).
User steps	Open incident → inspect resources → assign → confirm dispatch details → send → monitor acknowledgement and status.
Internal processing	Check prerequisites; reserve the resource; create the dispatch order; transmit; await acknowledgement; start the configured timeout; advance state only on a confirmed event.
Agency and team actions	**Disaster Management staff record the agency's acknowledgement, rejection with reason, or progress report, or a trusted integration reports it.** Limited to assigned dispatches. **Agencies have no login.**
Application response	Show Requested / Waiting until acknowledgement; rejected, failed and escalated states remain explicit.
Database and status changes	Resource assignment and dispatch state recorded; conflict checks prevent double-booking; each state change stores the actor or the integration that reported it.

Aspect	Execution detail
Notifications	The receiving team or agency contact gets the request; the operator gets the acknowledgement or failure status; a **staff member holding the supervisor permission** is notified on timeout or escalation.
Success	The receiving team's confirmation is recorded by **Disaster Management staff or a trusted integration**, and the timestamp and the recorder are stored.
Completion	Dispatch completion is set by a Disaster Management operator or a trusted integration, never inferred.
State definitions	**Acknowledged** = the receiving side has confirmed receipt of the order. **Dispatched** = the resource has been released and is en route.
Failure	No resource, transmission failure, rejection or timeout → a non-success state; release, reassign, escalate or retry per policy.
Flow	Verified incident → assign resource → create order → transmit → Requested → Acknowledged → Dispatched → Completed, or Rejected / Escalated / Failed.
Cross-references	DM-FR-09, DM-FR-10, DM-FR-08, DM-FR-20 *(optional)*, AM-FR-03, DM-NFR-13.

7.6 Registration, login and role assignment

Aspect	Execution detail
Purpose	Create and authenticate accounts while keeping the three fixed roles intact.
Preconditions	Public sign-up is enabled; the request is within its rate limit.
User steps	Sign up or log in → verify contact where required → reach the dashboard permitted by the role.

Aspect	Execution detail
Internal processing	**Public sign-up assigns the Normal User role and nothing else.** Staff and admin accounts are created only under AM-FR-02, with MFA required. Role and permission claims are resolved server-side on every request, never from the client.
Application response	Generic failure messages that do not reveal whether an account exists.
Database and status changes	Account, role, permission set and session records; failed attempts recorded for abuse control.
Success	A session whose effective role matches the stored role.
Failure	Invalid credentials, MFA failure, suspended account, rate limit reached, or a request claiming a role that does not exist (EH-11).
Cross-references	NU-FR-01, DM-FR-01, AM-FR-01, AM-FR-02, **AM-FR-11**.

7.7 Evidence and content moderation

Aspect	Execution detail
Purpose	Ensure user-supplied text and files are never rendered unsafely and that evidence is reviewed before wider visibility.
Preconditions	Evidence uploaded; the moderation policy is active (AM-FR-12); a staff member holds the moderator permission.
User steps	Open the moderation queue → inspect the item → approve, quarantine or remove → record the reason.
Internal processing	Scan on upload; auto-quarantine per policy; queue for human decision; store the decision with actor, time and reason; escape or sanitise all user text on render.
Application response	Quarantined items show a placeholder and reason to staff and are never shown publicly.

Aspect	Execution detail
Database and status changes	Moderation state Pending / Approved / Quarantined / Removed with full history.
Success	Every publicly visible item carries an approved moderation state.
Failure	Scanner outage → items stay Pending, not Approved; backlog beyond the threshold raises an alert.
Cross-references	NU-FR-05, DM-FR-18, AM-FR-12, NU-NFR-13, DM-NFR-14.

7.8 Incident closure

Aspect	Execution detail
Purpose	End an incident's operational life with a recorded reason, so no record is left in an indefinite open state.
Preconditions	The staff member holds the closure permission; dependent dispatches and allocations are resolved or explicitly overridden.
User steps	Open incident → review open actions → select a closure reason → add an optional public summary → close.
Internal processing	Validate dependencies; set Closed with reason, actor and time; freeze operational edits; keep the record readable and auditable.
Application response	The closure reason is visible to staff and to the reporter; reopening requires the same permission and is itself audited.
Database and status changes	Status = Closed; closure reason, actor and timestamp stored; linked records marked resolved.
Success	Closed with an auditable reason.
Failure	Open dependent actions without an override; missing reason; missing permission.
Cross-references	DM-FR-15, NU-FR-06, DM-NFR-06.

7.9 Safety content authoring and approval

Aspect	Execution detail
Purpose	Publish safety guidance that citizens can rely on, **versioned per scenario**.
Preconditions	Draft exists; a staff member holds the Publisher/Approver permission.
User steps	Draft → attach source → submit for approval → approve → publish.

Aspect	Execution detail
Internal processing	Every version is retained; the author and the approver must be recorded; only approved versions are served to Normal Users.
Application response	The public page shows the official status, source and last update time.
Database and status changes	Content version records with state Draft / Approved / Published / Superseded.
Success	The approved version is published and attributable.
Failure	Missing permission, unapproved draft, or a superseded version still being served.
Cross-references	DM-FR-17, DM-FR-11, NU-FR-08, DM-NFR-15.

7.10 Abuse and rate-limit control

Aspect	Execution detail
Purpose	Keep public endpoints usable during an emergency despite spam and automated abuse.
Preconditions	Thresholds configured under AM-FR-11; monitoring active.
User steps	None. The control is applied automatically; an affected user sees a clear message and a retry path.
Internal processing	Count requests per account, per IP and per endpoint; throttle, challenge or block; record every decision; alert Application Management on sustained abuse.
Application response	State that a limit was reached without revealing the threshold, and point to an alternative route for a genuine emergency.
Database and status changes	Abuse records, block entries and counters, all auditable.
Success	Abuse is contained and legitimate reporting remains possible.

Aspect	Execution detail
Failure	Thresholds too strict block genuine reports; distributed abuse stays under the threshold; the control service itself fails, in which case it fails safe and alerts.
Cross-references	AM-FR-11, NU-NFR-12, DM-NFR-14, AM-NFR-13, EH-18, EH-19.

7.11 Data export or deletion request *(Optional)*

Aspect	Execution detail
Purpose	Let a user obtain or remove their personal data within the limits of retention obligations.
Preconditions	NU-FR-14 and AM-FR-13 are adopted; the requester is authenticated.
User steps	Request export or deletion → confirm identity → await the outcome.
Internal processing	Queue; verify identity; apply retention and legal holds; execute or refuse with a recorded reason.
Application response	Show clearly what was exported, what was removed and what had to be retained, and why.
Success	Request resolved and recorded.
Failure	Identity unverified, retention obligation prevents deletion, or export generation fails (EH-26).
Cross-references	NU-FR-14, AM-FR-13, NU-NFR-15, AM-NFR-16.

8. Non-Functional Requirements – Normal User

Numerical thresholds are intentionally not invented. Any measurable target is a **proposed target until approved by [named approver]. 13 core requirements; 15 with the optional items.**

ID	Name	Explanation	Importance	How it can be tested
NU-NFR-01	Usability	Forms and status flows should be clear, consistent and easy to learn. **The three supported scenarios are described clearly on the report form**, so a user can tell at once whether their situation fits.	High	Scenario-based task tests and usability review.
NU-NFR-02	Accessibility	Keyboard access, readable contrast, labels, focus order and accessible status messages are supported across the website.	High	Accessibility audit and assistive-technology tests.
NU-NFR-03	Performance	Common user actions provide timely feedback; numerical targets are proposed only after workload analysis.	High	Load and response-time tests against approved targets.
NU-NFR-04	Availability	Core functions remain usable during normal service periods, with graceful degradation when a dependency fails.	High	Availability and dependency-outage tests.

ID	Name	Explanation	Importance	How it can be tested
NU-NFR-05	Privacy	Profile, location, evidence and contact data are restricted to **authorised Disaster Management staff only**. Agencies receive only what staff choose to share in dispatch messages. *(If NU-FR-14 is adopted, users may also request an export or deletion.)*	Critical	RBAC and privacy tests, plus penetration testing.
NU-NFR-06	Compatibility	Supported modern browsers and desktop and mobile form factors are documented and tested.	Medium	Compatibility matrix.
NU-NFR-07	Payment security	Sensitive payment credentials are handled by secure gateway mechanisms; any application storage follows the approved design.	Critical	Security review, gateway tests and penetration testing.
NU-NFR-08	Transaction reliability	Retries, callbacks, timeouts and duplicate attempts preserve the correct donation state.	Critical	Fault injection, idempotency and reconciliation tests.
NU-NFR-09	Transparency	Users can distinguish submitted, moderated, verified, pending, confirmed, closed and out-of-scope states.	High	State-transition and UI tests.
NU-NFR-10	Donor privacy	Donor identity and visibility follow approved privacy rules and least privilege.	High	Privacy and access tests.

ID	Name	Explanation	Importance	How it can be tested
NU-NFR-11	Timeliness	Incident acknowledgements and updates are designed for rapid feedback; targets require approval.	High	End-to-end timing tests.
NU-NFR-12	Abuse resistance	Spam and automated submissions are contained without blocking genuine emergency reporting (**AM-FR-11**).	Critical	Rate-limit, bot-simulation and false-positive tests.
NU-NFR-13	Content safety	User text and uploaded files are never rendered unsafely; evidence is moderated before wider visibility (pairs with DM-FR-18).	Critical	Injection and upload-safety tests; moderation workflow tests.
NU-NFR-14	Offline draft *(Optional)*	An unsent report draft is saved in the browser and shown in a clear "Not yet sent" state.	Medium	Browser-storage and retry tests.
NU-NFR-15	Data retention and user rights *(Optional)*	Users can request an export or deletion of their data within retention limits. Depends on NU-FR-14.	High	Data-request workflow and retention tests.

9. Non-Functional Requirements – Disaster Management

Any measurable target is a **proposed target until approved by [named approver]. 15 core requirements; 16 with the optional item.**

ID	Name	Explanation	Importance	How it can be tested
DM-NFR-01	Usability	Dashboards minimise cognitive load and foreground urgent and pending actions, including the moderation queue and overdue dispatches.	High	Expert review and task tests.
DM-NFR-02	Accessibility	Operational screens meet the selected accessibility baseline where applicable.	Medium	Accessibility audit.
DM-NFR-03	Performance	Search, dashboard and map operations remain responsive under approved operational load.	High	Load and stress tests.
DM-NFR-04	Availability	Core operations degrade safely when non-critical integrations fail.	Critical	Dependency outage and failover tests.
DM-NFR-05	Role-based access	Only authorised staff can moderate, verify, prioritise, dispatch, publish, close or access restricted data. **Only three roles exist.** Publisher/Approver, closure, moderation and supervisor are separate permissions inside Disaster Management. **No self-escalation.**	Critical	RBAC matrix and penetration tests.

ID	Name	Explanation	Importance	How it can be tested
DM-NFR-06	Auditability	Critical decisions, including closures and moderation outcomes, are attributable to a human operator with time and rationale where required.	Critical	Audit completeness and integrity tests.
DM-NFR-07	Interoperability	Dispatch, map and notification integrations use documented interfaces and state mappings. **Agency acknowledge, reject and progress states, whether staff-recorded or reported by a trusted integration, map to internal dispatch states.**	High	API contract and integration tests.
DM-NFR-08	Resilience	Loss of AI, map, notification or agency services must not corrupt records or falsely advance state.	Critical	Chaos and fault-injection tests.
DM-NFR-09	AI explainability	Recommendations show understandable factors, uncertainty and version metadata where feasible. **A suggestion outside the three scenarios is flagged Out of Scope.**	Critical	Explainability review against representative cases.
DM-NFR-10	Human oversight	Critical decisions cannot be finalised solely by AI.	Critical	Workflow enforcement and negative tests.

ID	Name	Explanation	Importance	How it can be tested
DM-NFR-11	Data integrity	Concurrent updates preserve incident, dispatch, resource and aid consistency.	Critical	Concurrency and transaction tests.
DM-NFR-12	Analytics transparency	Metrics identify definitions, freshness and the treatment of pending and unknown records. **Analytics cover only the three scenarios; out-of-scope and unknown records are counted separately.**	High	Metric reconciliation tests.
DM-NFR-13	Dispatch timeliness and escalation	An unacknowledged dispatch reaches a configured timeout and escalates to a staff member holding the supervisor permission, rather than waiting indefinitely (DM-FR-09, AM-FR-03).	Critical	Timeout, escalation and notification tests.
DM-NFR-14	Abuse and moderation handling	Staff can triage flagged and rate-limited submissions, and the moderation backlog stays within its configured threshold (DM-FR-18, **AM-FR-11**, **AM-FR-12**).	High	Queue-load, backlog-alert and moderation workflow tests.
DM-NFR-15	Content authority	Only approved, versioned content is published publicly, and every published item is attributable to its author and approver (DM-FR-11, DM-FR-17).	Critical	Publishing permission and version-integrity tests.

ID	Name	Explanation	Importance	How it can be tested
DM-NFR-16	Operational continuity *(Optional)*	Documented manual fallbacks let staff continue verification, dispatch recording and closure during an extended integration outage.	High	Tabletop exercise and degraded-mode drills.

10. Non-Functional Requirements – Application Management

Any measurable target is a **proposed target until approved by [named approver]. 14 core requirements; 16 with the optional items.**

ID	Name	Explanation	Importance	How it can be tested
AM-NFR-01	Maintainability	Services and configuration are modular, documented and testable. The configuration surface covers scenarios (enable/disable only), notification settings, timeouts, escalation, duplicate detection, abuse thresholds and moderation policy.	High	Architecture, code and change-impact review.
AM-NFR-02	Scalability	Growth in users, incidents, evidence and contributions is supported without redesigning core workflows.	High	Capacity and load tests using approved assumptions.

ID	Name	Explanation	Importance	How it can be tested
AM-NFR-03	Monitoring	Critical services and integrations emit actionable telemetry and alerts, including the moderation backlog and the agency/dispatch integration.	Critical	Monitoring coverage and alert simulations.
AM-NFR-04	Data integrity	Administrative changes use validation, controlled transactions and audit records.	Critical	Mutation and rollback tests.
AM-NFR-05	Backup and recovery	Backups are protected, verifiable and restorable under an approved recovery plan.	Critical	Restore drills and integrity tests.
AM-NFR-06	Security	Admin interfaces use strong authentication, least privilege, secure sessions and protected secrets. **Staff and admin accounts require MFA. Only three roles exist. The role list and the scenario list are fixed, and adding to either needs a formal change. Critical changes need a second approver (AM-FR-14, if adopted).**	Critical	Security tests, secret scanning and penetration testing.
AM-NFR-07	Payment integration monitoring	Gateway health, callback authenticity, reconciliation mismatches and pending transactions are observable.	Critical	Outage, callback and reconciliation simulations.

ID	Name	Explanation	Importance	How it can be tested
AM-NFR-08	Notification reliability	Failed notifications retry safely and remain visible without duplicate spam, with the in-app status as the fallback source of truth. *(Opt-out handling applies only if NU-FR-07 keeps it.)*	High	Provider-outage and duplicate-prevention tests.
AM-NFR-09	Auditability	Admin, security and configuration actions are retained and protected from unauthorised alteration, including **staff-recorded agency acknowledgements**.	Critical	Audit access and integrity tests.
AM-NFR-10	Compatibility and interoperability	Supported platform and external interface versions are documented, including the terms of any agency messaging or dispatch integration contract.	Medium	Compatibility and contract tests.
AM-NFR-11	Compliance	Deployment follows applicable data-protection, payment, accessibility, retention and security obligations.	Critical	Compliance evidence checklist and review.
AM-NFR-12	Resilience	Component failure triggers safe degradation, recovery or operator escalation rather than silent data loss.	Critical	Failure-injection and recovery tests.

ID	Name	Explanation	Importance	How it can be tested
AM-NFR-13	Abuse resistance	Rate limits and abuse controls hold under load without suppressing genuine emergency traffic. Depends on **AM-FR-11**.	Critical	Attack simulation and false-positive measurement.
AM-NFR-14	Configuration governance	Scenario, role, timeout and moderation settings are versioned, attributable and reversible. Depends on **AM-FR-03** and, if adopted, **AM-FR-14**.	Critical	Configuration audit, rollback and approval tests.
AM-NFR-15	Traceability *(Optional)*	Every FR and NFR traces to a design component and at least one test case.	High	Traceability matrix review.
AM-NFR-16	Data lifecycle *(Optional)*	Retention, archival and deletion schedules are defined per data class and enforced. Depends on AM-FR-13.	High	Retention-policy and deletion-job tests.

11. Application Error Handling and Recovery

The system preserves the last confirmed state and records failures. It never infers success merely because a request was sent. Recovery is explicit and auditable. **25 core cases; 26 with the optional item.**

ID	Error	Trigger	Application response	Recovery path
EH-01	Missing fields	Validation fails	Highlight the fields; preserve safe input; create no incomplete success. *(A browser draft is offered only if NU-NFR-14 is adopted.)*	Correct and resubmit.
EH-02	Invalid login	Authentication rejects	Generic failure; rate-limit or lockout as configured under **AM-FR-11**; do not leak whether the account exists.	Retry or account recovery.
EH-03	Unsupported upload	Type, size or security check fails	Reject or quarantine; explain the allowed category.	Upload approved evidence.
EH-04	Duplicate report	Similarity rules identify a candidate	Do not auto-delete; show matches for human review.	Link, merge or retain by policy.
EH-05	Unverified incident	Evidence insufficient	Keep the non-verified state; request information; no false dispatch.	Reporter provides evidence (NU-FR-11); staff review.
EH-06	AI uncertainty	Low confidence, contradiction or outage	Mark uncertain or unavailable; continue manual triage.	Human assessment.
EH-07	Failed donation	Gateway declines	Set Failed; allow a safe retry; issue no receipt.	Retry with an idempotent attempt.

ID	Error	Trigger	Application response	Recovery path
EH-08	Pending donation	Timeout or unknown result	Keep Pending; prevent duplicate charging; reconcile later.	Callback or reconciliation.
EH-09	Gateway outage	Payment service unavailable	No false success; show temporary unavailability; alert monitoring.	Retry after recovery, or an approved alternate.
EH-10	Invalid or unavailable aid	Unsuitable, or no matching need	Non-success state; explain the next action.	Modify, withdraw or reallocate.
EH-11	Unauthorised access	Protected action requested	**Access outside a role's permissions, or a request that claims a role that does not exist, is denied and logged.** No protected data leaks.	Authorised role or permission grant; security review.
EH-12	Server or network failure	Request interrupted	Recoverable error; safe retry with idempotency; never assume success. *(Draft preservation applies only if NU-NFR-14 is adopted.)*	Retry and reconcile.
EH-13	Failed notification	Provider rejects or fails	Record the failure; apply the retry policy; the in-app status is the source of truth. *(Opt-out handling applies only if opt-out is adopted.)*	Provider recovery or alternate channel.

ID	Error	Trigger	Application response	Recovery path
EH-14	Failed aid delivery	Pickup, transport or receipt fails	Keep the non-success state; record the reason; allow reschedule or reallocation.	New delivery attempt, or close with reason.
EH-15	Dispatch timeout	No acknowledgement within the configured window	Keep Requested; mark Escalated; notify a staff member holding the supervisor permission.	Resend, reassign or escalate (DM-NFR-13).
EH-16	Dispatch rejected	The receiving team or agency declines	Record Rejected with the stated reason and the recorder; never show dispatch success.	Reassign the resource, or select another recipient.
EH-17	Late or conflicting acknowledgement	An acknowledgement arrives after escalation, or contradicts the stored state	Keep both events in history; the later authorised event wins; flag the conflict for staff. **The acknowledgement is recorded by staff or a trusted integration**, with the recorder stored.	Staff reconcile the dispatch state.
EH-18	Submission flood	Report, flag or donation volume exceeds the threshold (**AM-FR-11**)	Throttle or challenge; state that a limit was reached without revealing the threshold; alert monitoring.	Retry after the window, or use the stated emergency alternative.

ID	Error	Trigger	Application response	Recovery path
EH-19	Automated abuse	Repeated failed logins or bot-like patterns (**AM-FR-11**)	Challenge or block the source; log the decision; do not reveal account existence.	Unblock after review; adjust thresholds.
EH-20	Unsafe content detected	Scan or moderation identifies malicious or harmful content	Quarantine immediately; never render it; alert moderators and Application Management.	Moderator decision; permanent removal with reason.
EH-21	Moderation backlog	Pending moderation exceeds its configured threshold	Keep items Pending, never auto-approve; raise an alert; show the delay to staff.	Add moderator capacity; adjust policy (DM-FR-18, **AM-FR-12**).
EH-22	Unauthorised publish	A staff member without the Publisher/Approver permission attempts to publish	Deny and log; keep the content in Draft.	Route to an approver (DM-FR-11, DM-FR-17).
EH-23	Invalid closure	Closure attempted with open dependencies or no reason	Refuse; list the open dispatches and allocations; require a reason or an explicit override with rationale.	Resolve dependencies, or close with a recorded override.
EH-24	Rejected configuration change	An attempt to add a role or a fourth scenario, or an invalid value	Refuse; keep the previous version active; log the attempt.	Submit a formal change; use AM-FR-14 approval if adopted.

ID	Error	Trigger	Application response	Recovery path
EH-25	Unsupported scenario	The submitted scenario is not one of the three, or the AI suggests another type	**The server rejects the submission, or flags the AI suggestion as Out of Scope.**	Pick one of the three. If none fits, the report is not accepted and the site shows a pointer to local emergency services.
EH-26	Data request failure *(Optional)*	Export or deletion cannot be completed (**AM-FR-13**)	Explain what was retained and why; never report a deletion that did not happen.	Retry, or refuse with a recorded legal or retention reason.

Consistency notes. EH-02, EH-18 and EH-19 tie to **AM-FR-11**. EH-21 ties to DM-FR-18 and **AM-FR-12**. EH-26 ties to **AM-FR-13**. EH-01 and EH-12 carry draft wording only if NU-NFR-14 is adopted; EH-13 carries opt-out wording only if opt-out is adopted.

12. End-to-End Operational Workflow

Moderation sits between intake and verification, so the steps are lettered A to L. **No agency actor appears in the workflow**; agency involvement happens through Disaster Management under step G. **12 steps.**

Report → validate and store → moderate → verify → AI assist → prioritise → coordinate resources → dispatch → publish verified updates → close → audit and analytics. Donation and aid may start independently and rejoin at allocation.

Step	Stage	System behaviour
A	Intake	A Normal User submits a report. The system validates the location and details, and **the scenario must be one of the three, validated on the server**. A Submitted record is created.
B	Moderation	Evidence and user text are scanned and reviewed under the active moderation policy. Nothing unapproved is rendered beyond the reviewing staff.

Step	Stage	System behaviour
C	Verification	Disaster Management reviews the evidence and decides Verified, Rejected or Needs Information.
D	AI assistance	AI suggests classification, urgency and duplicates. Staff review and record the final decision. A suggestion outside the three scenarios is flagged Out of Scope.
E	Prioritisation	Authorised staff assign a priority and rationale; the change is auditable.
F	Resources	Staff check availability and reserve suitable resources. They **coordinate with external agencies through Disaster Management staff (DM-FR-10); agencies have no login**.
G	Dispatch	The request is sent. State advances only after a confirmed event, and the **acknowledgement is recorded by staff or a trusted integration**. Timeout triggers escalation.
H	Updates	Verified public updates are **published by staff holding the Publisher/Approver permission**, with timestamps and source status.
I	Contributions	Money and physical aid can be contributed independently. Payment is reconciled; physical aid is verified.
J	Allocation and delivery	Verified aid is matched and allocated against a need from DM-FR-19 where adopted, or **against needs entered by Disaster Management staff (DM-FR-13)**, then tracked. Delivered requires confirmation.
K	Closure	A staff member holding the closure permission closes the incident with a recorded reason, or reopens it under the same permission.
L	Monitoring	Application Management monitors services, logs, security, backups, abuse controls and integrations; analytics are reconciled.

Re-lettering note. Because moderation was inserted after intake, references elsewhere shift: AI assistance is step **D** (was C), and dispatch is step **G** (was F). Section 7.5 and scenario 13.5 refer to step G.

13. Detailed Scenario Examples

All three supported disasters appear across these examples. **11 core scenarios; 13 with the optional ones.**

13.1 Urban Flooding incident. A user reports flooding at a road junction, adds a location pin and a photo, and notes stranded people. The application stores the report as Submitted and displays an ID. The photo enters moderation and is approved. Staff review the evidence. AI suggests Urban Flooding and highlights stranded-person indicators with uncertainty. Staff accept the classification, set priority and check resources. A dispatch request is sent; until acknowledgement the interface stays Requested/Waiting. Only after a recorded confirmation does the dispatch become Acknowledged and then Dispatched. A verified public update may then be published by a staff member holding the Publisher/Approver permission.

13.2 Industrial Fire with AI uncertainty. A report contains limited smoke evidence. AI confidence is low and a nearby report appears similar. The system labels the recommendation uncertain and shows candidate duplicates without deleting anything. Staff inspect the evidence and make the authoritative verification and priority decision. If verified, operational processing continues; otherwise the record remains non-verified.

13.3 Monetary donation. A user selects an approved campaign, limited to the three supported scenarios, and enters an amount. The application creates Pending and hands the payment to the secure gateway. Gateway confirmation plus callback and reference validation changes the transaction to Confirmed and triggers the receipt. A decline becomes Failed. A timeout or ambiguous response stays Pending for reconciliation. No success receipt is issued before confirmation.

13.4 Physical aid. A contributor registers 50 sealed water cartons. The system creates Pending Verification. Staff verify suitability, allocate the aid to an approved need and create a delivery task. Pickup and transit events are recorded. If delivery fails, the state remains non-success and can be rescheduled or reallocated. Only an authorised receipt confirmation creates Delivered.

13.5 Emergency dispatch. For a verified high-priority Industrial Fire, staff select an available resource at step G. The application checks for conflicts, creates a dispatch order and sends it to the receiving team. If the acknowledgement is recorded, the state advances. If communication fails or the team rejects the request, the application shows the failure or waiting state and allows a safe retry or reassignment. It never claims dispatch success.

13.6 Integration failure. If the gateway is unavailable, the user receives a temporary-unavailability message and no success receipt. Application Management sees an integration alert and pending transactions remain traceable for reconciliation. If the

agency messaging or dispatch integration is out, staff record acknowledgements manually and the dispatch state still advances only on a recorded confirmation. If the server fails during incident submission, the retry path uses idempotency and status checking so no duplicate incident is created or falsely reported as successful. *(If the browser draft is adopted, an unsent report stays in a "Not yet sent" state.)*

13.7 Needs Information — Cyclone. A user files a **Cyclone** report but the affected area is unclear and the shelter need is not stated. Staff set the incident to Needs Information with a specific question. The user answers under NU-FR-11, the incident returns to the review queue, and staff then verify it and set a priority. Nothing is dispatched while the incident is in Needs Information.

13.8 Agency acknowledgement. A dispatch is sent to an external agency contact. **The acknowledgement arrives either through a staff entry or through a trusted integration.** The stored record names the recorder and the timestamp. If nothing arrives within the configured window, the dispatch escalates to a staff member holding the supervisor permission.

13.9 Abuse and rate limiting. A script submits hundreds of near-identical reports from one source. The abuse controls throttle and then block that source, the duplicates are surfaced for human review rather than deleted, and monitoring alerts Application Management. Genuine reports from other users continue to be accepted throughout.

13.10 Incident closure. An Urban Flooding incident has had its dispatch completed and its aid delivered. A staff member holding the closure permission verifies there are no open dependencies, selects a closure reason and closes the incident. The reporter sees the closed state and the reason in their history. The record stays readable and auditable, and reopening is itself audited.

13.11 Report outside the three scenarios. A user wants to report an earthquake. The form offers no such option. If a tampered request reaches the server with another scenario value, **the server rejects it (EH-25)** and the site shows a pointer to local emergency services. Nothing is stored as an incident.

13.12 Evidence moderation *(Optional)*. An uploaded image is auto-quarantined by the scanner. It is never rendered publicly. A moderator reviews it, removes it with a recorded reason, and the incident continues with the remaining evidence.

13.13 Safety content or public update approval *(Optional)*. A staff member drafts updated Cyclone shelter guidance. Publication is refused until a holder of the Publisher/Approver permission approves it. The previous version stays live until the new version is approved, and both versions are retained.

14. Stakeholder Responsibilities, Summary and Open Decisions

Stakeholder	Core responsibilities	System support
Normal User	Report incidents; supply location, details and evidence; track reports; respond to information requests; withdraw or update own reports; retrieve receipts; receive alerts and safety information; donate money; contribute physical aid.	Guided forms with a fixed three-scenario choice, transparent states, secure gateway flow, contribution tracking and status history.
Disaster Management	Moderate evidence; verify; AI-assisted triage; prioritise; map; coordinate resources; dispatch and handle timeouts and escalation; record agency acknowledgements; close incidents; manage campaigns; author and approve safety content; manage needs; publish verified updates; oversee and allocate aid; analytics.	Operational dashboard, moderation queue, evidence review, AI advisory panel, resource and dispatch controls, agency messaging, closure workflow and analytics.
Application Management	Accounts, roles and permissions; configuration; abuse and rate-limit controls; moderation configuration; enforcing the three fixed roles; monitoring; audit; data; backup and recovery; notifications; payment integration.	Admin console, RBAC, abuse controls, moderation policy, audit logs, telemetry, backup and recovery, and integration monitoring.

There is **no Agency/Partner row**. Agencies are external parties reached through Disaster Management.

System summary

The Disaster Response Coordination Hub creates a controlled digital workflow for Cyclone, Industrial Fire and Urban Flooding. It connects citizen reports and contributions to authorised operational decisions while preserving human accountability. **Only three roles and three disaster scenarios are supported, and agencies are external parties with no login.** AI improves triage support but does not replace authorised judgement. Donations and physical aid can begin independently of incident reporting. Payment, dispatch, notification and delivery outcomes require confirmation. **Every failure path ends in an explicit safe state rather than an assumed success, and every incident ends**

either in a resolved operational state or as Closed with a recorded reason. Application Management provides the security, auditability, resilience, monitoring and recovery foundation needed for dependable disaster response.

Final totals

Item	Original	Core	Full
Functional requirements	34	42 (NU 12, DM 18, AM 12)	49 (NU 15, DM 20, AM 14)
Non-functional requirements	35	42 (NU 13, DM 15, AM 14)	47 (NU 15, DM 16, AM 16)
Error cases	14	25	26
Execution specifications	5	10	11
Scenario examples	6	11	13
Workflow steps	10	12	12
Roles	3	3	3

Open decisions

Four decisions remain with the team. They change only the optional counts, not the core baseline.

1. Keep or skip DM-FR-19 (needs management). If skipped, staff enter needs directly under DM-FR-13, and Section 12 step J follows that route.
2. Keep or skip the browser draft (NU-FR-13 and NU-NFR-14). This also decides the draft wording in EH-01, EH-12 and scenario 13.6.
3. Name the approver for every *proposed target*, replacing **[named approver]** in Sections 8, 9 and 10.
4. Keep or skip the optional data-request items (NU-FR-14, AM-FR-13, NU-NFR-15, AM-NFR-16, EH-26, specification 7.11).

Implementation note

Before implementation, the team should baseline these requirements, define proposed measurable performance and availability targets, document supported browsers and API contracts, define the incident, dispatch, donation and aid state machines, identify

applicable legal and compliance obligations for the deployment jurisdiction, and trace every FR and NFR to design components and test cases (**AM-NFR-14** governs configuration, and AM-NFR-15 covers traceability if adopted).

15. Route Specification

Every functional requirement in Sections 4, 5 and 6 maps to at least one route. Routes are grouped by role so the RBAC boundary is visible in the URL itself: `/api/v1/...` is public or Normal User, `/api/v1/dm/...` requires the Disaster Management role, and `/api/v1/admin/...` requires the Application Management role.

15.1 Conventions

Rule	Specification
Base path	`/api/v1`. The version is part of the path, so a breaking change ships as `/api/v2`.
Naming	Lower-case, hyphenated, plural resource nouns. Path parameters are `:id`. A state change that is not a simple field update is a sub-resource, for example `POST /dispatches/:id/acknowledgement`, never `POST /dispatch-acknowledge`.
Methods	`GET` read, `POST` create or record an event, `PATCH` partial update, `PUT` full replace of a settings object, `DELETE` remove. State transitions use `POST` on a sub-resource so each one is auditable.
Authentication	Session cookie or bearer token. `/dm/` and `/admin/` routes additionally require MFA (DM-FR-01, AM-FR-01).
Authorisation	The role and permission set are resolved server-side on **every** request. A request claiming a role that does not exist, or acting outside its permissions, returns `403` and is logged (EH-11).
Scenario validation	Any route accepting a `scenario` value accepts only `cyclone`, `industrial_fire`, `urban_flooding`. Anything else returns `422` with `UNSUPPORTED_SCENARIO` (EH-25).

Rule	Specification
Idempotency	`POST /incidents`, `POST /donations` and `POST /aid-contributions` require an `Idempotency-Key` header, so a retry after a network failure cannot create a duplicate (EH-12, NU-FR-13).
Rate limiting	All public and write routes pass through the abuse controls (AM-FR-11) and may return `429` (EH-18, EH-19).
Pagination	List routes take `?page=` and `?limit=`, and filter routes take `?scenario=`, `?status=`, `?from=`, `?to=`.
Ownership	A Normal User route returns only records owned by the session user. A mismatch returns `404`, not `403`, so record existence is not leaked.

15.2 Page routes (website)

Route	Page	Access
`/`	Landing page with safety information and published updates	Public
`/register`	Sign-up; creates a Normal User only	Public
`/login`	Login for all three roles; MFA step for staff and admin	Public
`/safety` and `/safety/:scenario`	Approved safety guidance per scenario	Public
`/updates` and `/updates/:id`	Verified public updates	Public
`/report`	Incident report form with the fixed three-scenario choice	Normal User
`/incidents/:id`	Own incident status, timeline, evidence and closure reason	Normal User

Route	Page	Access
`/account` and `/account/notifications`	Profile, history, receipts, notification preferences	Normal User
`/donate` and `/donate/:campaignId`	Campaign list and donation form	Normal User
`/contribute-aid`	Physical aid contribution form	Normal User
`/dm`	Operational dashboard	Disaster Management
`/dm/incidents/:id`	Review, triage, priority, dispatch and closure workspace	Disaster Management
`/dm/moderation`	Moderation queue	Disaster Management, moderator permission
`/dm/resources`, `/dm/map`, `/dm/aid`, `/dm/analytics`	Resource, map, aid and analytics workspaces	Disaster Management
`/dm/content` and `/dm/campaigns`	Safety content and campaign management	Disaster Management, Publisher/Approver for publishing
`/admin`	Admin console	Application Management
`/admin/users`, `/admin/config`, `/admin/abuse`, `/admin/moderation`, `/admin/audit`, `/admin/backups`, `/admin/integrations`	Administration workspaces	Application Management

Any page route requested without the matching role redirects to `/login` when unauthenticated, and returns the 403 page when authenticated but unauthorised.

15.3 Normal User and public API routes

Method	Route	Purpose	FR
`POST`	`/api/v1/auth/register`	Create an account. Assigns the Normal User role only.	NU-FR-01

Method	Route	Purpose	FR
`POST`	`/api/v1/auth/login`	Authenticate. Returns an MFA challenge for staff and admin.	NU-FR-01, DM-FR-01, AM-FR-01
`POST`	`/api/v1/auth/mfa/verify`	Complete the MFA challenge.	DM-FR-01, AM-FR-01
`POST`	`/api/v1/auth/logout`	End the session.	NU-FR-01
`POST`	`/api/v1/auth/password-reset`	Request a reset link. Generic response either way.	NU-FR-01
`GET`	`/api/v1/me`	Read own profile.	NU-FR-02
`PATCH`	`/api/v1/me`	Update permitted profile fields. Role changes are rejected.	NU-FR-02
`GET`	`/api/v1/scenarios`	The three supported scenarios, for the report form.	NU-FR-03
`POST`	`/api/v1/incidents`	Submit an incident. Server-validates the scenario.	NU-FR-03
`GET`	`/api/v1/incidents/:id`	Read an own incident.	NU-FR-06
`PATCH`	`/api/v1/incidents/:id`	Correct an own incident before dispatch starts.	NU-FR-12

Method	Route	Purpose	FR
`POST`	`/api/v1/incidents/:id/withdraw`	Withdraw an own report with a reason.	NU-FR-12
`PUT`	`/api/v1/incidents/:id/location`	Set or correct the GPS or manual location.	NU-FR-04
`GET`	`/api/v1/geo/reverse-geocode`	Resolve coordinates to an address preview.	NU-FR-04
`POST`	`/api/v1/incidents/:id/evidence`	Upload evidence. Returns the moderation state.	NU-FR-05
`GET`	`/api/v1/incidents/:id/evidence`	List own evidence with per-file state.	NU-FR-05
`DELETE`	`/api/v1/incidents/:id/evidence/:evidenceId`	Remove own evidence before verification.	NU-FR-05
`GET`	`/api/v1/incidents/:id/timeline`	Status timeline, including closure reason.	NU-FR-06
`GET`	`/api/v1/me/incidents`	Own report history.	NU-FR-06
`GET`	`/api/v1/incidents/:id/information-requests`	Open staff questions on an own incident.	NU-FR-11
`POST`	`/api/v1/incidents/:id/information-requests/:requestId/response`	Answer a question and re-queue the incident.	NU-FR-11
`GET`	`/api/v1/me/notifications`	In-app notifications, the source of truth.	NU-FR-07
`PATCH`	`/api/v1/me/notifications/:id`	Mark read or dismissed.	NU-FR-07

Method	Route	Purpose	FR
`GET` `PUT`	`/api/v1/me/notification-preferences`	Read and set channel preferences and opt-out.	NU-FR-07
`GET`	`/api/v1/safety-content`	Approved safety guidance, filtered by `?scenario=`.	NU-FR-08
`GET`	`/api/v1/safety-content/:scenario`	The current approved version for one scenario.	NU-FR-08
`GET`	`/api/v1/public-updates`	Published verified updates.	DM-FR-11
`GET`	`/api/v1/campaigns`	Active, approved donation purposes.	NU-FR-09, DM-FR-16
`POST`	`/api/v1/donations`	Create a Pending transaction and get the gateway handoff.	NU-FR-09
`GET`	`/api/v1/donations/:id`	Exact payment state: Pending, Confirmed or Failed.	NU-FR-09
`GET`	`/api/v1/donations/:id/receipt`	Receipt. Returns `409` unless the transaction is Confirmed.	NU-FR-06, NU-FR-09
`GET`	`/api/v1/me/donations`	Own donation history.	NU-FR-06
`POST`	`/api/v1/aid-contributions`	Register physical aid.	NU-FR-10
`GET`	`/api/v1/aid-contributions/:id`	Verification, allocation and delivery state.	NU-FR-10

Method	Route	Purpose	FR
`GET`	`/api/v1/me/aid-contributions`	Own aid contribution history.	NU-FR-06, NU-FR-10
`POST`	`/api/v1/flags` *(Optional)*	Flag published or shared content.	NU-FR-15
`POST`	`/api/v1/me/data-requests` *(Optional)*	Request a data export or deletion.	NU-FR-14
`GET`	`/api/v1/me/data-requests/:id` *(Optional)*	Request outcome and what was retained.	NU-FR-14

The browser draft (NU-FR-13, optional) has **no route of its own**. It lives in browser storage and reuses `POST /api/v1/incidents` with the `Idempotency-Key` header on retry.

15.4 Disaster Management API routes

All routes below require the Disaster Management role. Routes marked with a permission additionally require that permission inside the role.

Method	Route	Purpose	Permission	FR
`GET`	`/api/v1/dm/dashboard`	Consolidated counts, queues and pending actions.	—	DM-FR-02
`GET`	`/api/v1/dm/incidents`	Incident queue with filters.	—	DM-FR-02
`GET`	`/api/v1/dm/incidents/:id`	Full incident record including evidence and history.	—	DM-FR-03
`POST`	`/api/v1/dm/incidents/:id/verification`	Record Verified, Rejected or Needs Information with rationale.	—	DM-FR-03
`POST`	`/api/v1/dm/incidents/:id/information-requests`	Ask the reporter for more information.	—	DM-FR-03

Method	Route	Purpose	Permission	FR
`GET`	`/api/v1/dm/incidents/:id/duplicates`	Duplicate candidates with rationale. Nothing is deleted.	—	DM-FR-04
`POST`	`/api/v1/dm/incidents/:id/duplicates/:candidateId/decision`	Link, merge or dismiss a candidate.	—	DM-FR-04
`POST`	`/api/v1/dm/incidents/:id/ai-triage`	Request an advisory recommendation.	—	DM-FR-05
`GET`	`/api/v1/dm/incidents/:id/ai-recommendations`	Stored recommendations, factors, uncertainty and model version.	—	DM-FR-05
`POST`	`/api/v1/dm/incidents/:id/ai-recommendations/:recId/decision`	Accept, modify or reject. Records the human decision separately.	—	DM-FR-05
`PUT`	`/api/v1/dm/incidents/:id/priority`	Set priority with rationale.	—	DM-FR-06
`GET`	`/api/v1/dm/map`	Operational map data with freshness metadata.	—	DM-FR-07
`GET`	`/api/v1/dm/map/layers`	Available layers and their sources.	—	DM-FR-07
`GET`	`/api/v1/dm/resources`	Resource inventory and availability.	—	DM-FR-08
`POST`	`/api/v1/dm/resources/:id/reservations`	Reserve a resource; conflict returns `409`.	—	DM-FR-08
`DELETE`	`/api/v1/dm/resources/:id/reservations/:reservationId`	Release a reservation.	—	DM-FR-08
`POST`	`/api/v1/dm/incidents/:id/dispatches`	Create and transmit a dispatch order.	dispatch	DM-FR-09

Method	Route	Purpose	Permission	FR
`GET`	`/api/v1/dm/dispatches`	Dispatch queue, including overdue and escalated.	—	DM-FR-09
`GET`	`/api/v1/dm/dispatches/:id`	Dispatch state and event history.	—	DM-FR-09
`POST`	`/api/v1/dm/dispatches/:id/acknowledgement`	**Staff record the agency's acknowledgement.** Stores recorder and timestamp.	dispatch	DM-FR-09, DM-FR-10
`POST`	`/api/v1/dm/dispatches/:id/rejection`	Record a rejection with the stated reason.	dispatch	DM-FR-09
`POST`	`/api/v1/dm/dispatches/:id/progress`	Record a progress report from the receiving side.	dispatch	DM-FR-09
`POST`	`/api/v1/dm/dispatches/:id/completion`	Mark the dispatch complete. Never inferred.	dispatch	DM-FR-09
`POST`	`/api/v1/dm/dispatches/:id/escalation`	Escalate after a timeout.	supervisor	DM-FR-09, DM-NFR-13
`POST`	`/api/v1/dm/incidents/:id/agency-messages`	Send a coordination message to external agency contacts.	—	DM-FR-10
`GET`	`/api/v1/dm/agency-messages/:id`	Sent, acknowledged, rejected or failed state and who recorded it.	—	DM-FR-10
`POST`	`/api/v1/dm/public-updates`	Draft a verified public update.	—	DM-FR-11
`POST`	`/api/v1/dm/public-updates/:id/publish`	Publish it. Returns `403` without the permission (EH-22).	publisher	DM-FR-11

Method	Route	Purpose	Permission	FR
`GET`	`/api/v1/dm/oversight/donations`	Confirmed money by campaign, with pending shown separately.	—	DM-FR-12
`GET`	`/api/v1/dm/oversight/aid`	Verified aid and its allocation status.	—	DM-FR-12
`POST`	`/api/v1/dm/aid-contributions/:id/verification`	Verify or reject registered aid.	—	DM-FR-13
`POST`	`/api/v1/dm/allocations`	Allocate verified aid to a need and create a delivery task.	—	DM-FR-13
`POST`	`/api/v1/dm/allocations/:id/delivery-events`	Record pickup, transit, delivered or failed.	—	DM-FR-13
`GET`	`/api/v1/dm/analytics`	Operational and contribution analytics. Out-of-scope records counted separately.	—	DM-FR-14
`POST`	`/api/v1/dm/incidents/:id/closure`	Close with a recorded reason. `409` if dependencies are open (EH-23).	closure	DM-FR-15
`POST`	`/api/v1/dm/incidents/:id/reopen`	Reopen a closed incident. Audited.	closure	DM-FR-15
`GET` `POST`	`/api/v1/dm/campaigns`	List and create campaigns. Scenario limited to the three.	—	DM-FR-16
`PATCH`	`/api/v1/dm/campaigns/:id`	Update a campaign.	—	DM-FR-16
`POST`	`/api/v1/dm/campaigns/:id/closure`	Close a campaign to new donations.	—	DM-FR-16

Method	Route	Purpose	Permission	FR
`GET` `POST`	`/api/v1/dm/safety-content`	List and draft safety content, versioned per scenario.	—	DM-FR-17
`GET`	`/api/v1/dm/safety-content/:id/versions`	Full version history with author and approver.	—	DM-FR-17
`POST`	`/api/v1/dm/safety-content/:id/approval`	Approve a draft version.	publisher	DM-FR-17
`POST`	`/api/v1/dm/safety-content/:id/publish`	Publish the approved version.	publisher	DM-FR-17
`GET`	`/api/v1/dm/moderation/queue`	Pending and quarantined items, with backlog age.	moderator	DM-FR-18
`POST`	`/api/v1/dm/moderation/:itemId/decision`	Approve, quarantine or remove with a reason.	moderator	DM-FR-18
`GET` `POST`	`/api/v1/dm/needs` *(Optional)*	List and create operational needs.	—	DM-FR-19
`PATCH`	`/api/v1/dm/needs/:id` *(Optional)*	Update quantity, priority or fulfilment.	—	DM-FR-19
`GET` `POST`	`/api/v1/dm/agency-contacts` *(Optional)*	Agency directory. Contacts only, **no accounts**.	—	DM-FR-20
`PATCH` `DELETE`	`/api/v1/dm/agency-contacts/:id` *(Optional)*	Maintain a contact.	—	DM-FR-20
`POST`	`/api/v1/dm/incidents/:id/agency-assignments` *(Optional)*	Assign a directory contact to an incident.	—	DM-FR-20

If DM-FR-19 is skipped, `POST /api/v1/dm/allocations` accepts an inline need object instead of a `needId`, matching the DM-FR-13 fallback.

15.5 Application Management API routes

All routes below require the Application Management role and MFA. There are **no agency account routes**, because agencies are an external integration.

Method	Route	Purpose	FR
`GET` `POST`	`/api/v1/admin/users`	List accounts; create a staff or admin account.	AM-FR-02
`GET`	`/api/v1/admin/users/:id`	Account detail with effective role and permissions.	AM-FR-02
`PUT`	`/api/v1/admin/users/:id/role`	Set the role. Only the three roles are accepted.	AM-FR-02
`PUT`	`/api/v1/admin/users/:id/permissions`	Grant or revoke publisher, moderator, closure, dispatch and supervisor permissions.	AM-FR-02
`POST`	`/api/v1/admin/users/:id/suspension`	Suspend an account.	AM-FR-02
`DELETE`	`/api/v1/admin/users/:id/suspension`	Reactivate an account.	AM-FR-02
`GET` `PUT`	`/api/v1/admin/config`	Read and update operational settings.	AM-FR-03
`PUT`	`/api/v1/admin/config/scenarios`	Enable or disable the three scenarios. Adding a fourth returns `422` (EH-24).	AM-FR-03

Method	Route	Purpose	FR
`PUT`	`/api/v1/admin/config/dispatch`	Dispatch timeout and escalation values.	AM-FR-03, DM-NFR-13
`PUT`	`/api/v1/admin/config/duplicate-detection`	Toggle and tune duplicate detection.	AM-FR-03
`GET`	`/api/v1/admin/config/versions`	Configuration version history with before and after values.	AM-FR-03, AM-NFR-14
`POST`	`/api/v1/admin/config/versions/:version/rollback`	Roll back to a previous version.	AM-FR-03
`GET`	`/api/v1/admin/health`	Service, queue and storage health.	AM-FR-04
`GET`	`/api/v1/admin/integrations`	Health of payment, notification, map and agency/dispatch integrations.	AM-FR-04
`GET`	`/api/v1/admin/audit-logs`	Search and filter audit records.	AM-FR-05
`GET`	`/api/v1/admin/data/tasks`	Data-quality and correction tasks.	AM-FR-06
`POST`	`/api/v1/admin/data/corrections`	Apply a controlled, traceable correction.	AM-FR-06

Method	Route	Purpose	FR
`GET` `PUT`	`/api/v1/admin/security/settings`	Security policy and session controls.	AM-FR-07
`POST`	`/api/v1/admin/security/secrets/:id/rotation`	Rotate a secret.	AM-FR-07
`GET` `POST`	`/api/v1/admin/backups`	List backups; trigger a protected backup.	AM-FR-08
`POST`	`/api/v1/admin/backups/:id/restore`	Restore with an integrity check.	AM-FR-08
`GET` `PUT`	`/api/v1/admin/notifications/config`	Channels, templates, routing and retry.	AM-FR-09
`POST`	`/api/v1/admin/notifications/test`	Send a test notification before activation.	AM-FR-09
`GET`	`/api/v1/admin/payments/health`	Gateway and callback health.	AM-FR-10
`GET`	`/api/v1/admin/payments/reconciliation`	Mismatches and long-pending transactions.	AM-FR-10
`GET` `PUT`	`/api/v1/admin/abuse/rules`	Rate-limit thresholds per endpoint.	AM-FR-11
`GET`	`/api/v1/admin/abuse/events`	Throttle, challenge and block decisions.	AM-FR-11
`GET` `POST`	`/api/v1/admin/abuse/blocks`	List and create blocks.	AM-FR-11
`DELETE`	`/api/v1/admin/abuse/blocks/:id`	Unblock after review.	AM-FR-11

Method	Route	Purpose	FR
`GET` `PUT`	`/api/v1/admin/moderation/policy`	File types, scan settings, auto-quarantine rules, backlog thresholds.	AM-FR-12
`GET`	`/api/v1/admin/data-requests` *(Optional)*	Export and deletion request queue.	AM-FR-13
`POST`	`/api/v1/admin/data-requests/:id/decision` *(Optional)*	Execute or refuse with a recorded reason.	AM-FR-13
`GET`	`/api/v1/admin/change-approvals` *(Optional)*	Critical changes pending a second approver.	AM-FR-14
`POST`	`/api/v1/admin/change-approvals/:id/approval` *(Optional)*	Approve or reject. Self-approval returns `403`.	AM-FR-14

15.6 Integration and webhook routes

These are machine-to-machine endpoints. They are authenticated by signature, never by a user session, and none of them grants an agency a login.

Method	Route	Purpose	FR
`POST`	`/api/v1/integrations/payments/callback`	Signed gateway callback. Verifies the signature, reconciles and sets Confirmed, Failed or Pending.	NU-FR-09, AM-FR-10
`POST`	`/api/v1/integrations/dispatch/acknowledgement`	A **trusted integration** reports an agency acknowledgement or rejection. Stores the integration as the recorder.	DM-FR-09, DM-FR-10

Method	Route	Purpose	FR
`POST`	`/api/v1/integrations/dispatch/progress`	A trusted integration reports progress or completion.	DM-FR-09
`POST`	`/api/v1/integrations/notifications/delivery-status`	Provider delivery receipt or failure.	NU-FR-07, AM-FR-09
`GET`	`/api/v1/health`	Liveness probe for monitoring.	AM-FR-04

15.7 Status codes and error mapping

Code	Meaning	Error cases
`200` / `201`	Success, or resource created. Never returned for an unconfirmed payment, dispatch or delivery.	—
`202`	Accepted and pending an external confirmation.	EH-08
`400`	Malformed request.	—
`401`	Not authenticated, or MFA incomplete.	EH-02
`403`	Authenticated but outside the role's permissions, or a non-existent role claimed.	EH-11, EH-22
`404`	Not found, or not owned by the requester.	EH-04, EH-05
`409`	State conflict: concurrent update, resource double-booking, receipt before confirmation, closure with open dependencies.	EH-14, EH-23
`413` / `415`	Upload too large, or an unsupported type.	EH-03
`422`	Validation failed, including an unsupported scenario or an attempt to add a role or scenario.	EH-01, EH-24, **EH-25**
`429`	Rate limit or abuse control triggered.	EH-18, EH-19

Code	Meaning	Error cases
`500`	Server failure. The client retries with the same `Idempotency-Key`.	EH-12
`503`	A dependency is unavailable: gateway, map, notification, AI or agency integration.	EH-06, EH-09, EH-13

Every non-`2xx` response carries a stable machine code (for example `UNSUPPORTED_SCENARIO`, `MODERATION_PENDING`, `DISPATCH_NOT_ACKNOWLEDGED`) so the interface can show the recovery path named in Section 11 rather than a raw status number.